IDDQ

IDDQ FAULT MODEL

IDDQ : Current drawn by a CMOS device in quiescent state (stable state – inputs are stable)

IDD -> Power supply current indicates the current flows from VDD to Gnd in a CMOS circuit.

Q -> quiescent

We apply certain test pattern and we wait for a certain amount of time so that the circuit becomes stable. After this we measure IDDQ (After manufacturing).

Idea of IDDQ testing :

- Good CMOS circuit, very low IDDQ
- Defective CMOS circuit, high IDDQ

<u>Types of defects detected by IDDQ testing :</u>

i. Stuck-on faults
ii. Bridging faults

If a chip has high IDDQ, it means that it is defective and it will fail in the early stages of its product life.

We can then use emission microscope to find out which logic is defective. It will generate a heat map and the defective logic will be shown as a highlighted region.

IDDQ FAULT MODEL

Example :

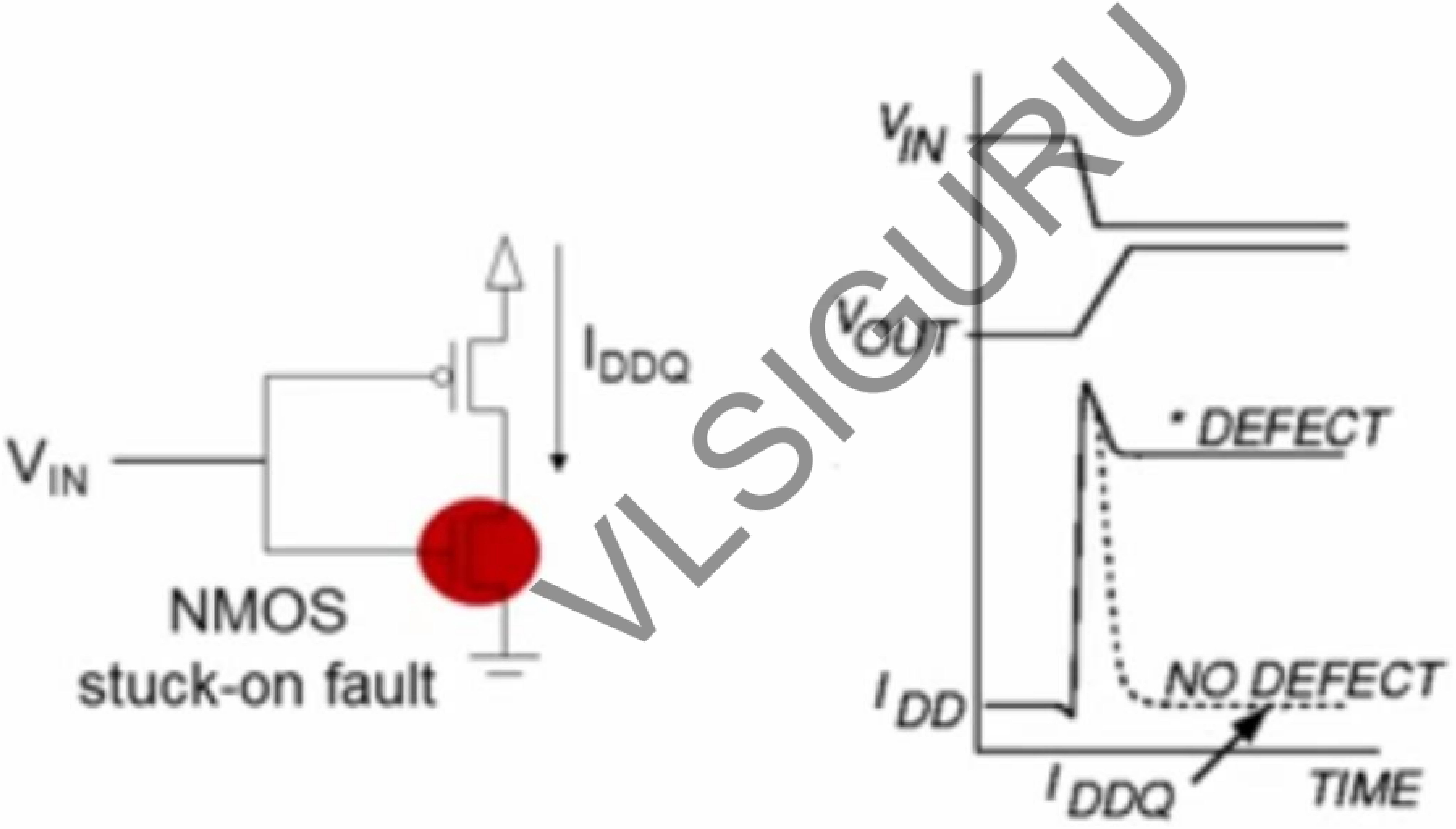
Stuck on fault :

Faulty transistor is always on.

Forms a conducting path between VDD and Gnd in static state (stable state)

IDDQ FAULT MODEL

Example :



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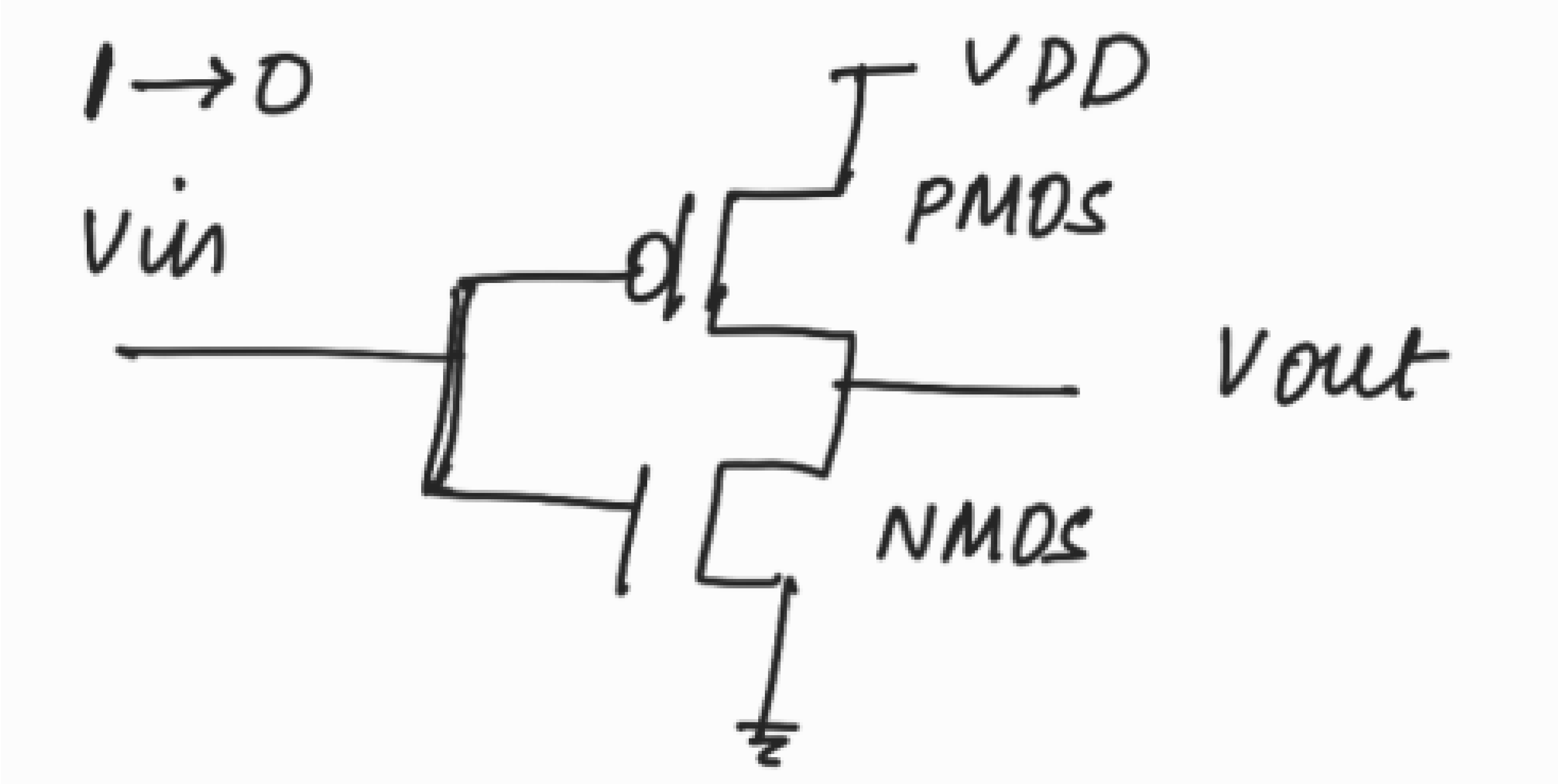
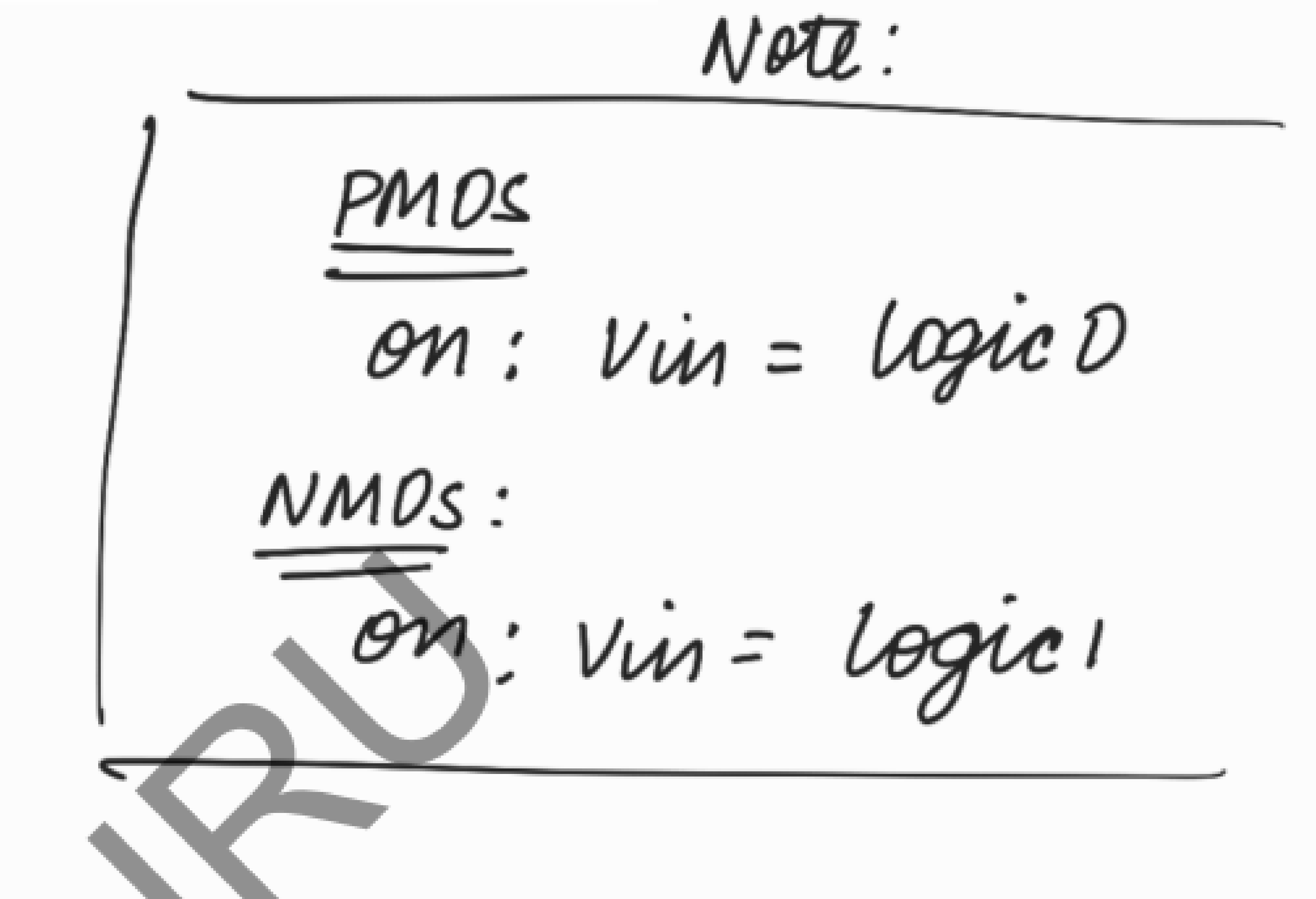
Good Circuit :



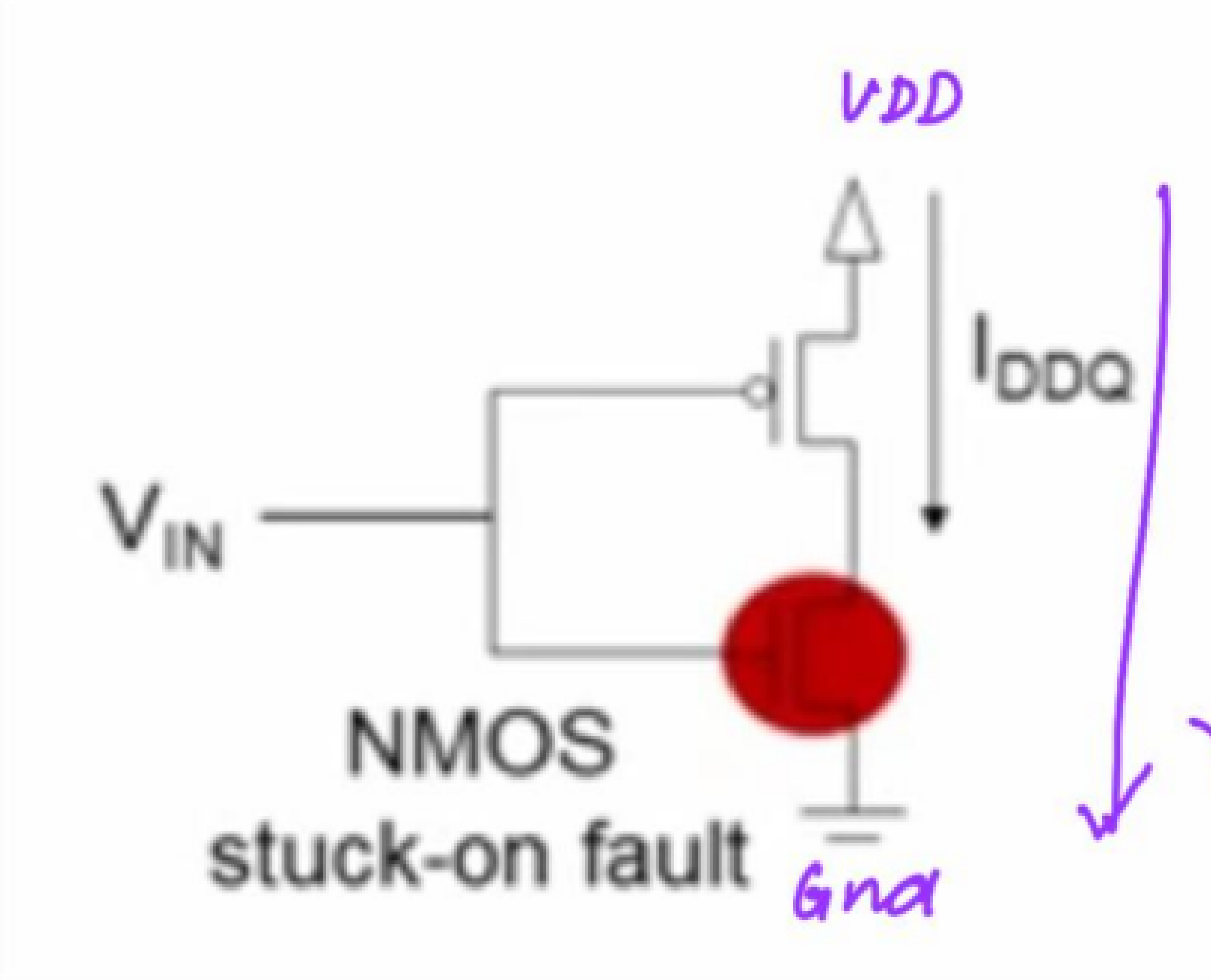


When the input voltage is changed from 5V (logic 1) to 0V (logic 0), initially NMOS will be on and PMOS will be off.

When input voltage is around 2.5 V (VDD/2) (if VDD = 5V), then at this time both NMOS and PMOS will be in their saturation region. So for a small amount of time both NMOS and PMOS will be conducting and hence during this time current flows from VDD to Gnd.

But as Vin reaches 0V (logic 0), PMOS will be on and NMOS will be off. Hence no current flows from VDD to Gnd.

But if the circuit has NMOS stuck on fault, we will observe a high IDDQ current.



even when Vin has reached 0V, NMOS will still be ON(stuck-on fault). This will occur when there is some manufacturing defect in NMOS.

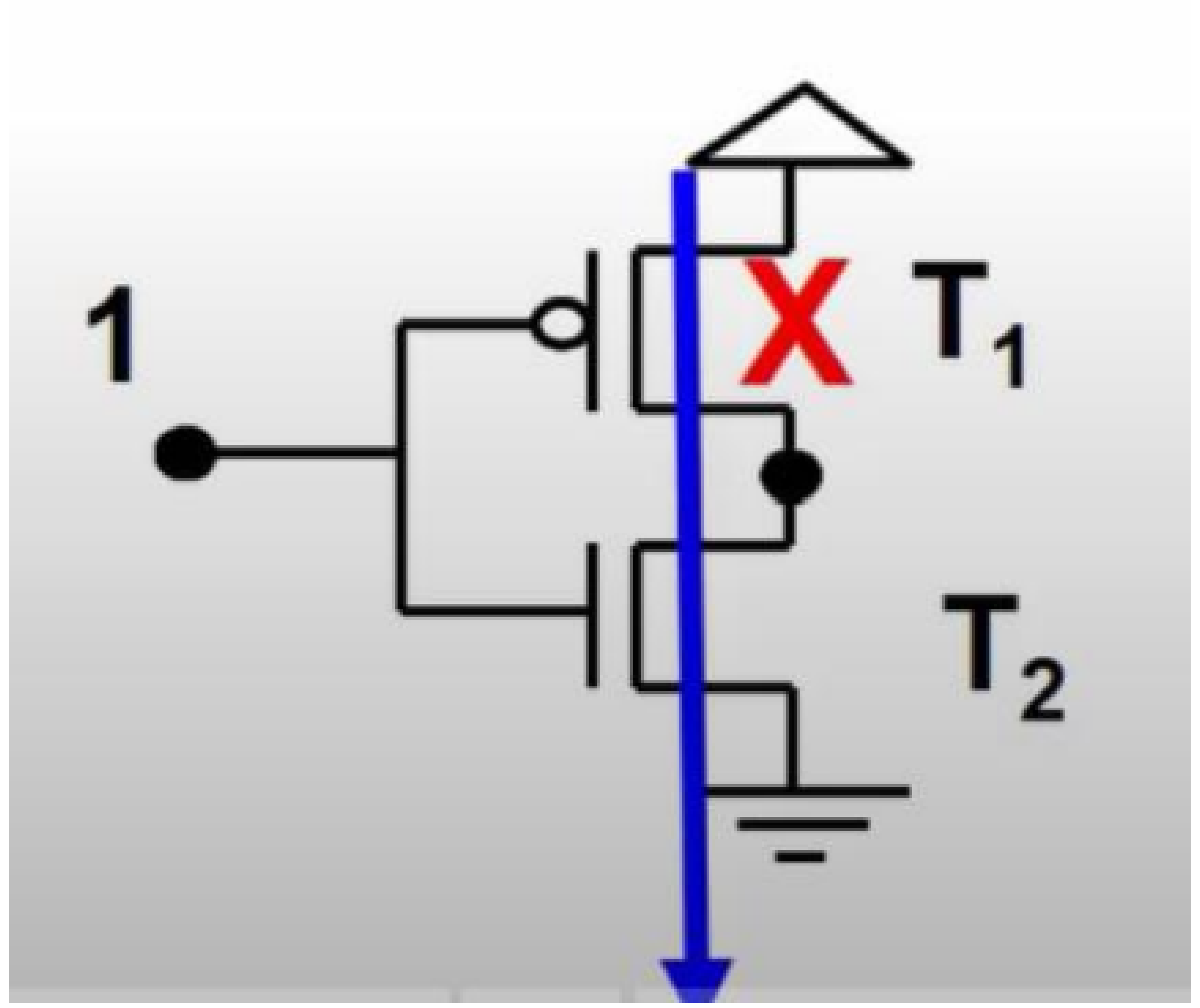
<u>resistive short</u>:

There is a direct path for the current to flow from VDD to Gnd. Hence current will be flowing continuously even after waiting for some time.

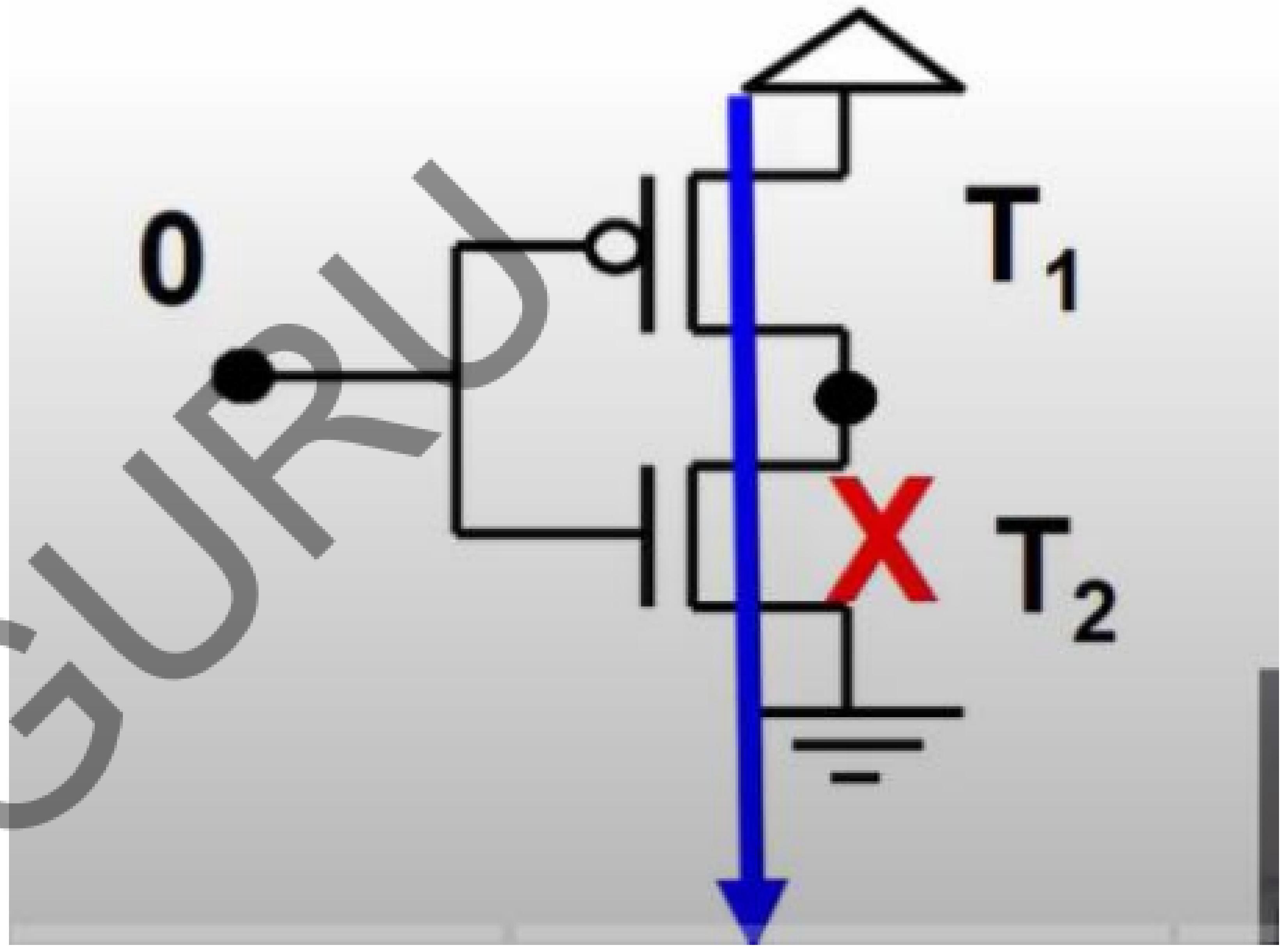
SUMMARY

When input is changed from logic 1 (5V) to logic 0 (0V), after a transition, a good circuit should consume no static current.

But if the circuit has NMOS stuck on fault, we will observe a high IDDQ current.



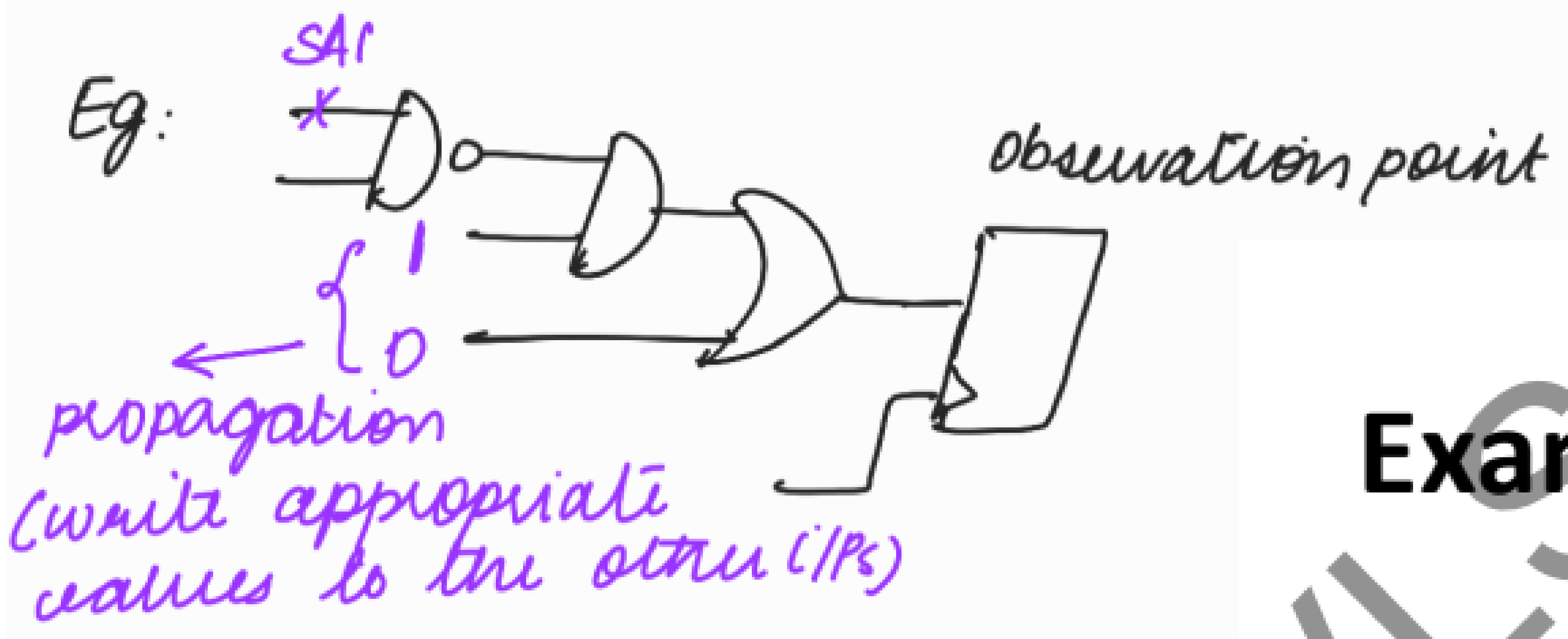
Suppose, this PMOS is stuck on. To detect this fault, we can apply logic 1



Suppose, this NMOS is stuck on. To detect this fault, we can apply logic 0

TEST PATTERNS FOR IDDQ

- Test set activate SA fault but effects need not propagate to any observation point. Propagation to observation point is not required because IDDQ can be measured by current.



Example :

For SA, in order to detect the fault in the NAND gate, the tool has to propagate it to the observation point.
But for IDDQ, tool will only try to activate the faults in the NAND gate. It will not try to propagate it to the observation point. Because in IDDQ we are not going to check the value at output. We are only going to check whether the cell draws excessive current in its stable state (quiescent state)

PATTERN LIST TO DETECT ALL STUCK ON FAULTS IN THE NAND GATE

A,B = {(1,1) ,(0,1), (1,0)}

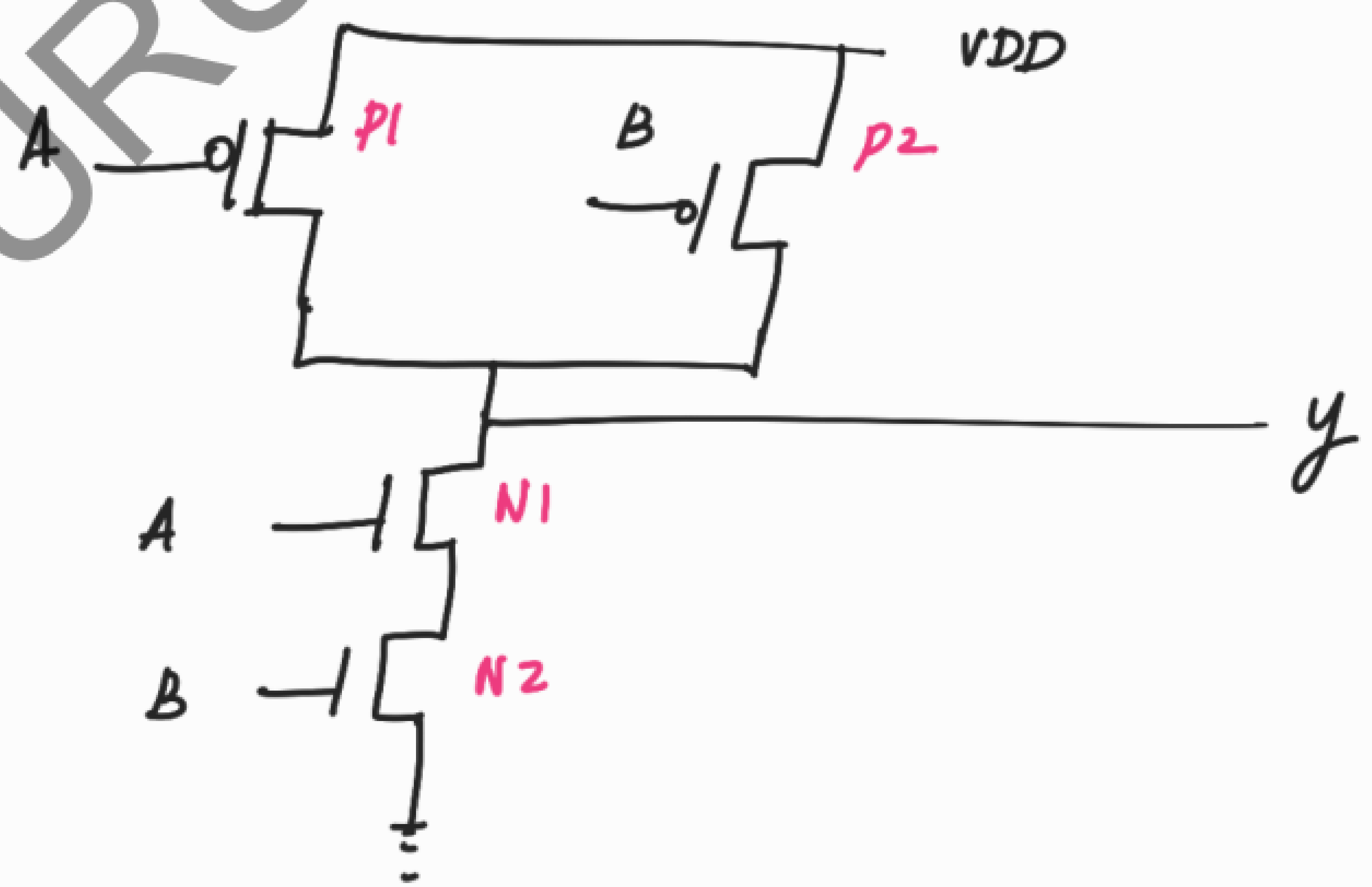
Explanation

NAND => $Y = \overline{A \cdot B}$

final eqn has complement

∴ don't add inverter at the end.

	·	+
pmos	‖l	series
nmos	series	‖l



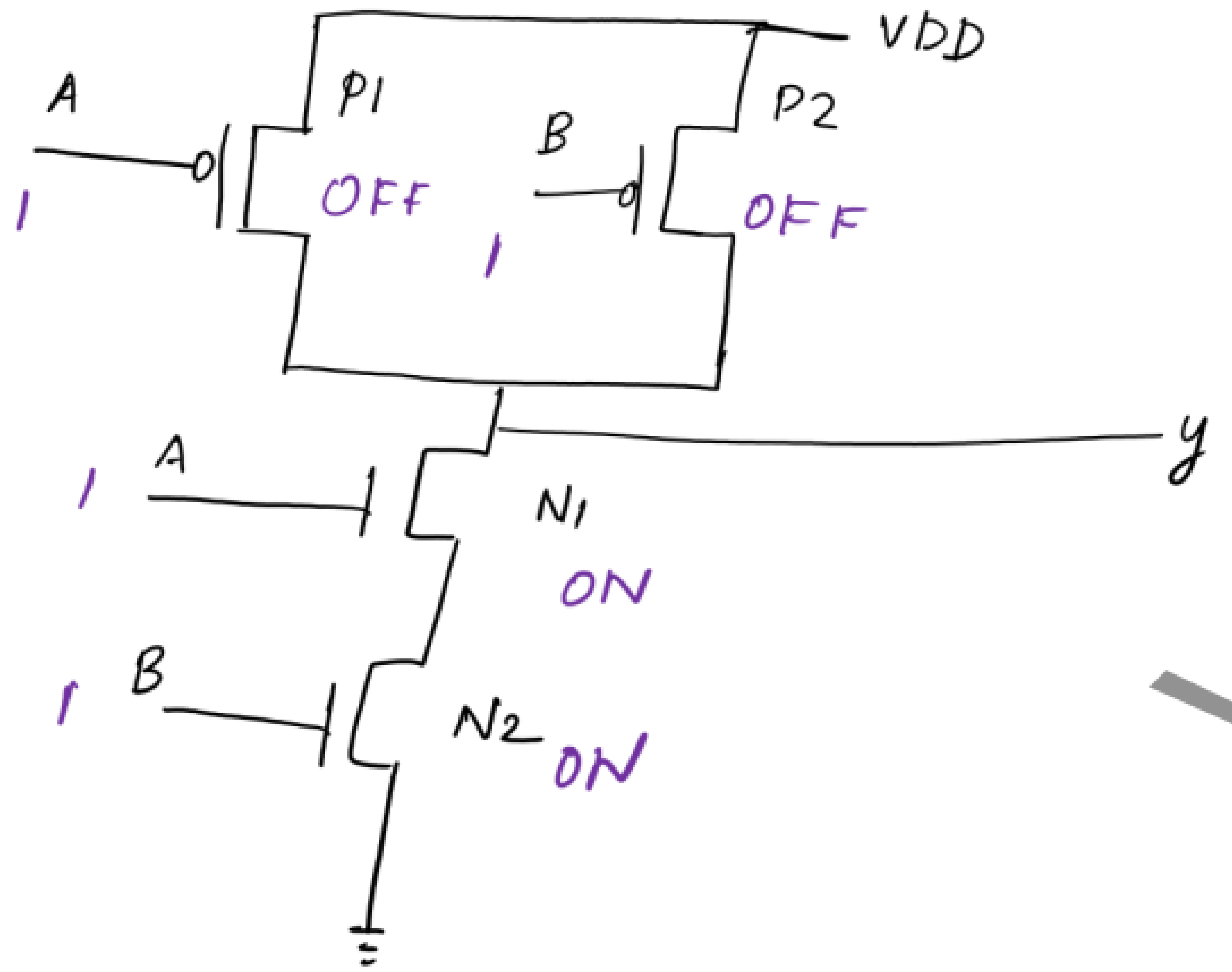
Pattern :- A = 1, B = 1

Note:

PMOS
on : Vin = logic 0

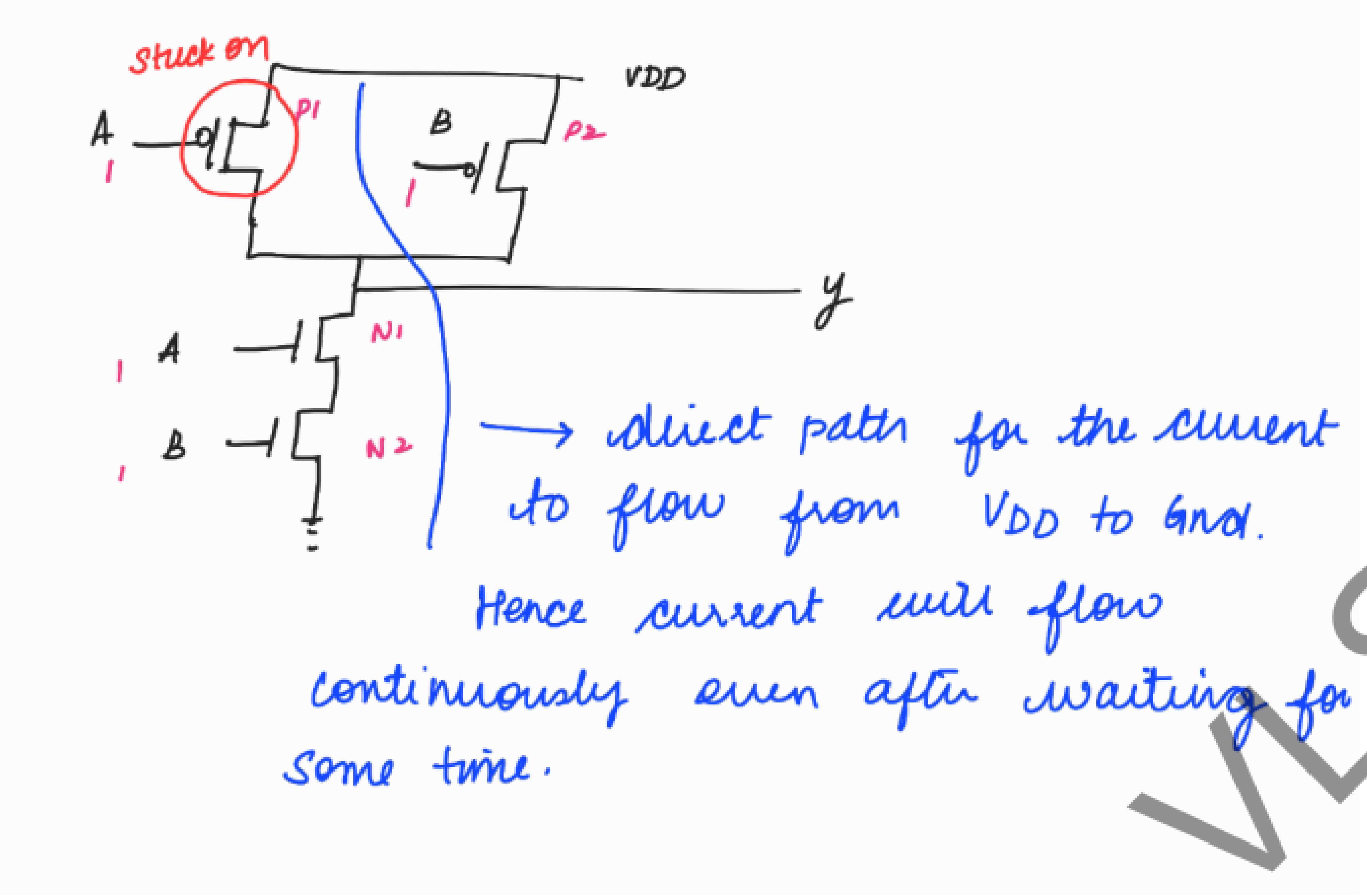
NMOS :
on : Vin = logic 1

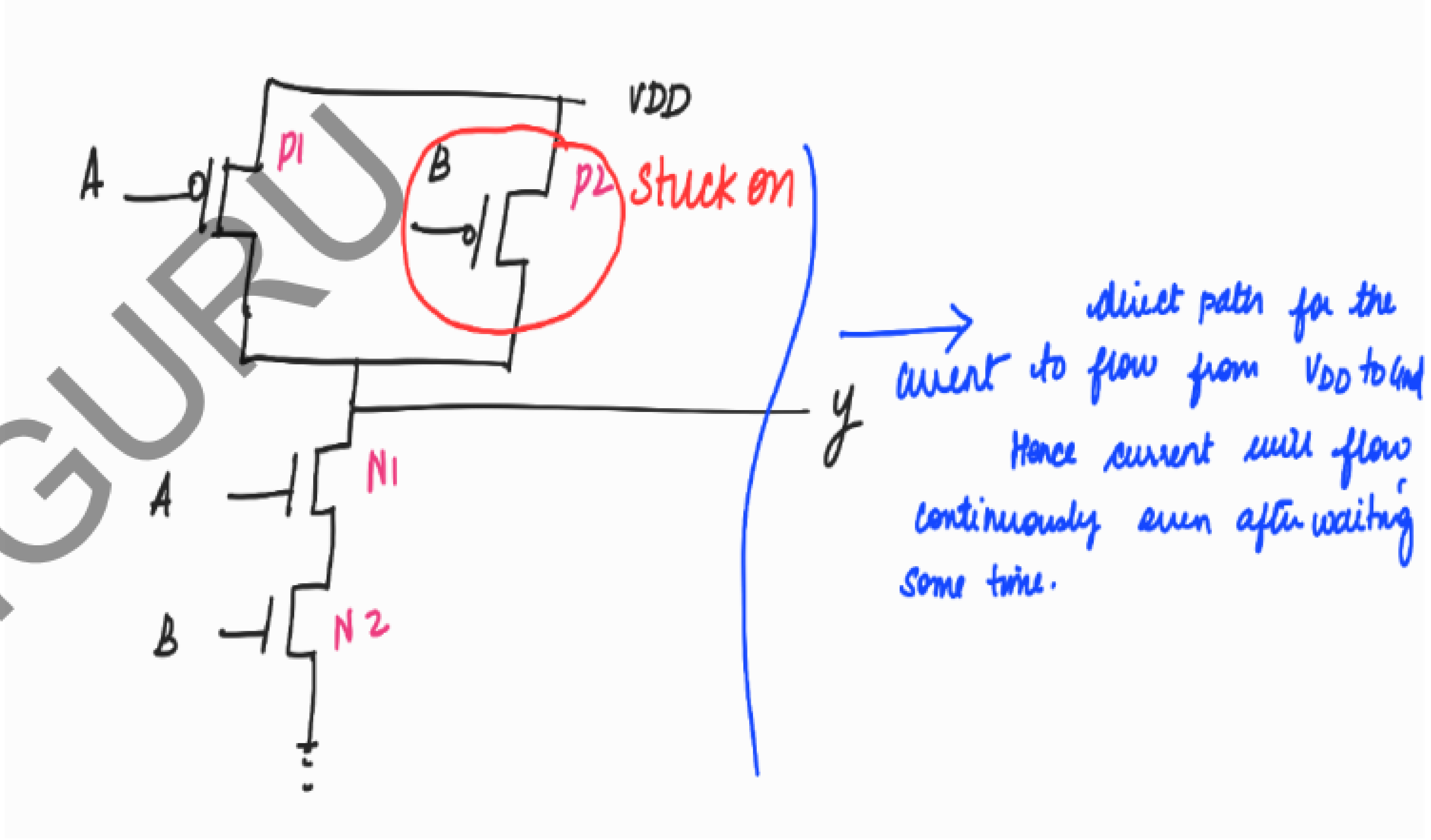
Good :-



There is no direct path for the current to flow from VDD to Gnd

This pattern will detect P1 stuck on and P2 stuck on faults.

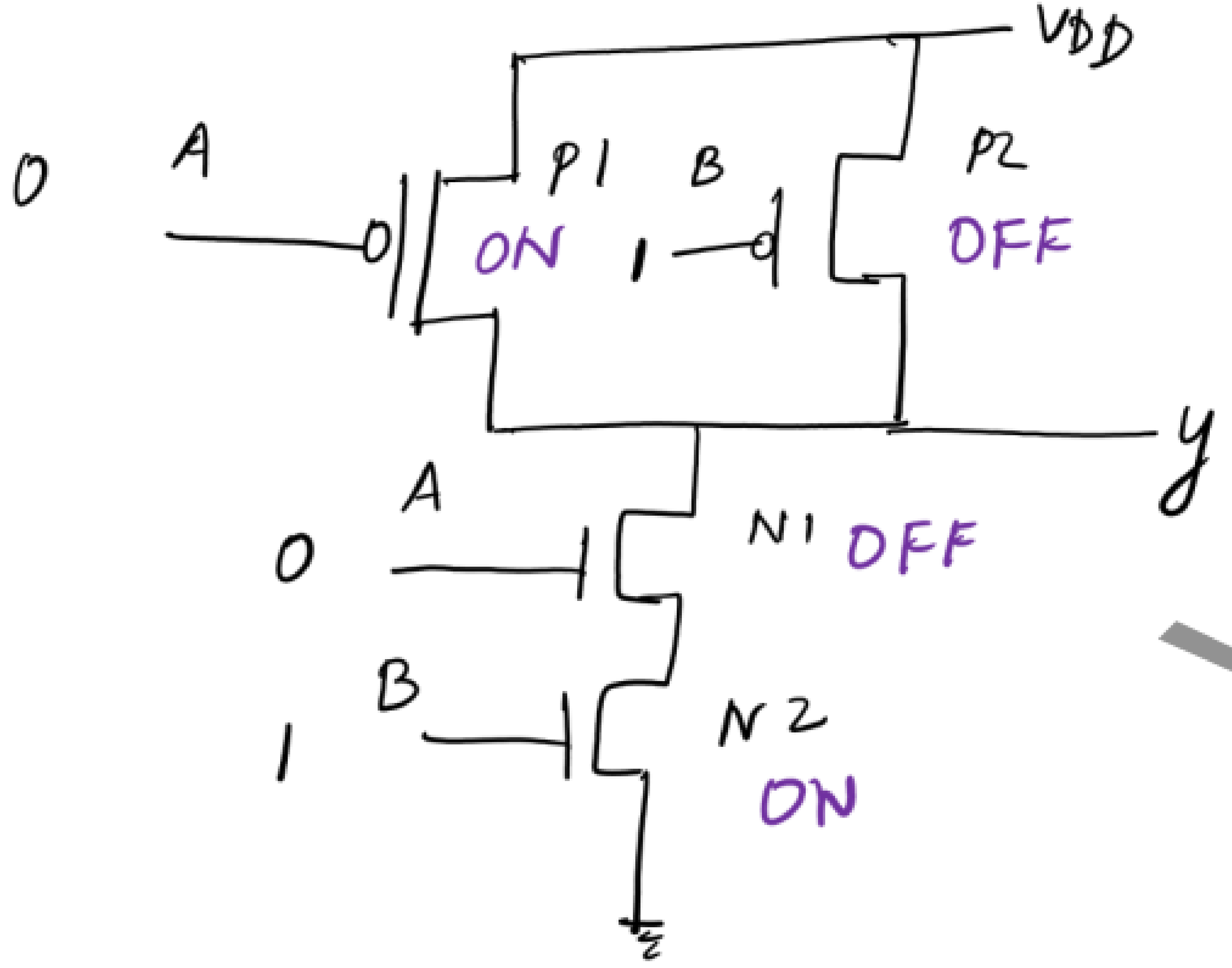




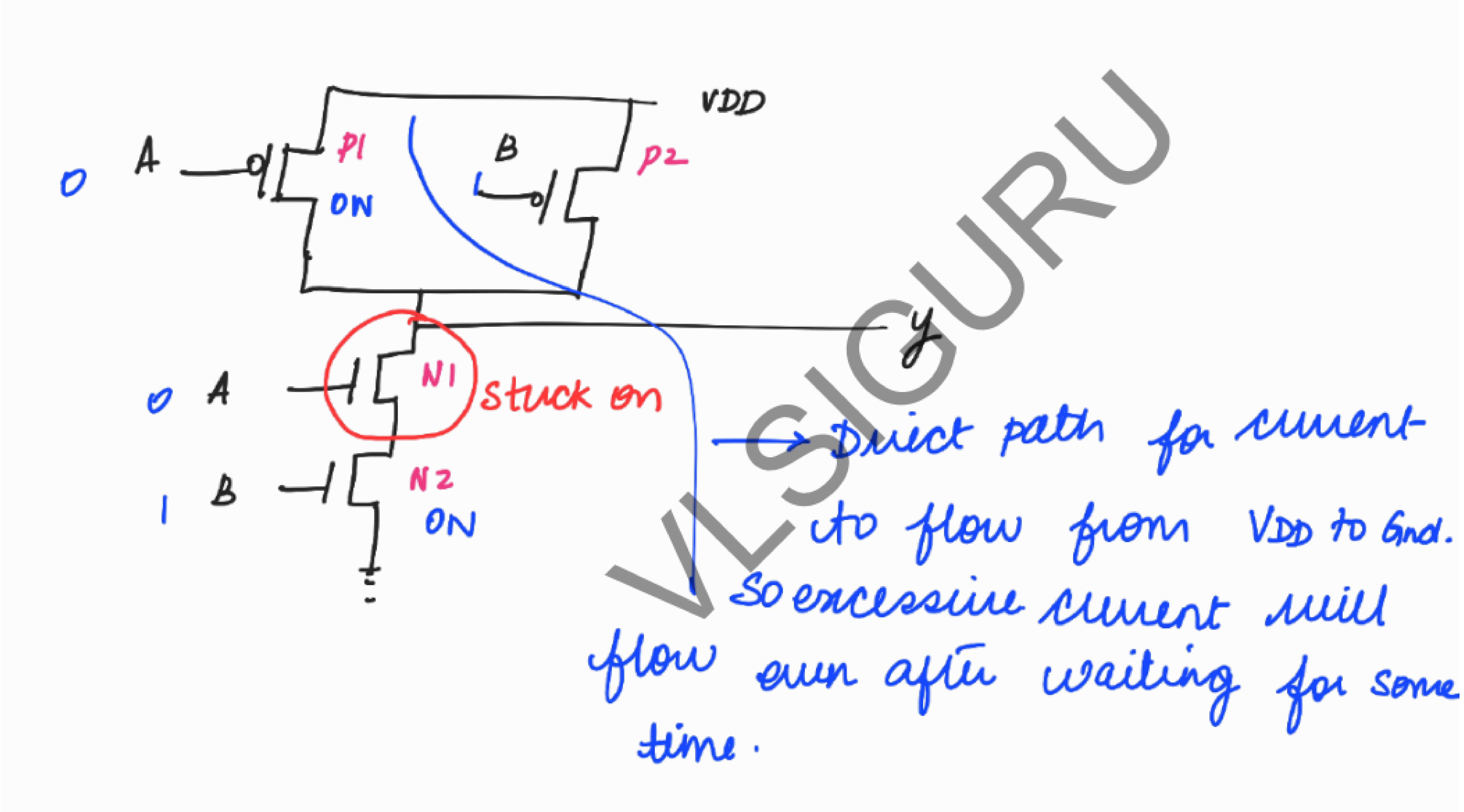
pattern A=0
B=1

Good :

no direct path for the current to flow from VDD to Gnd.



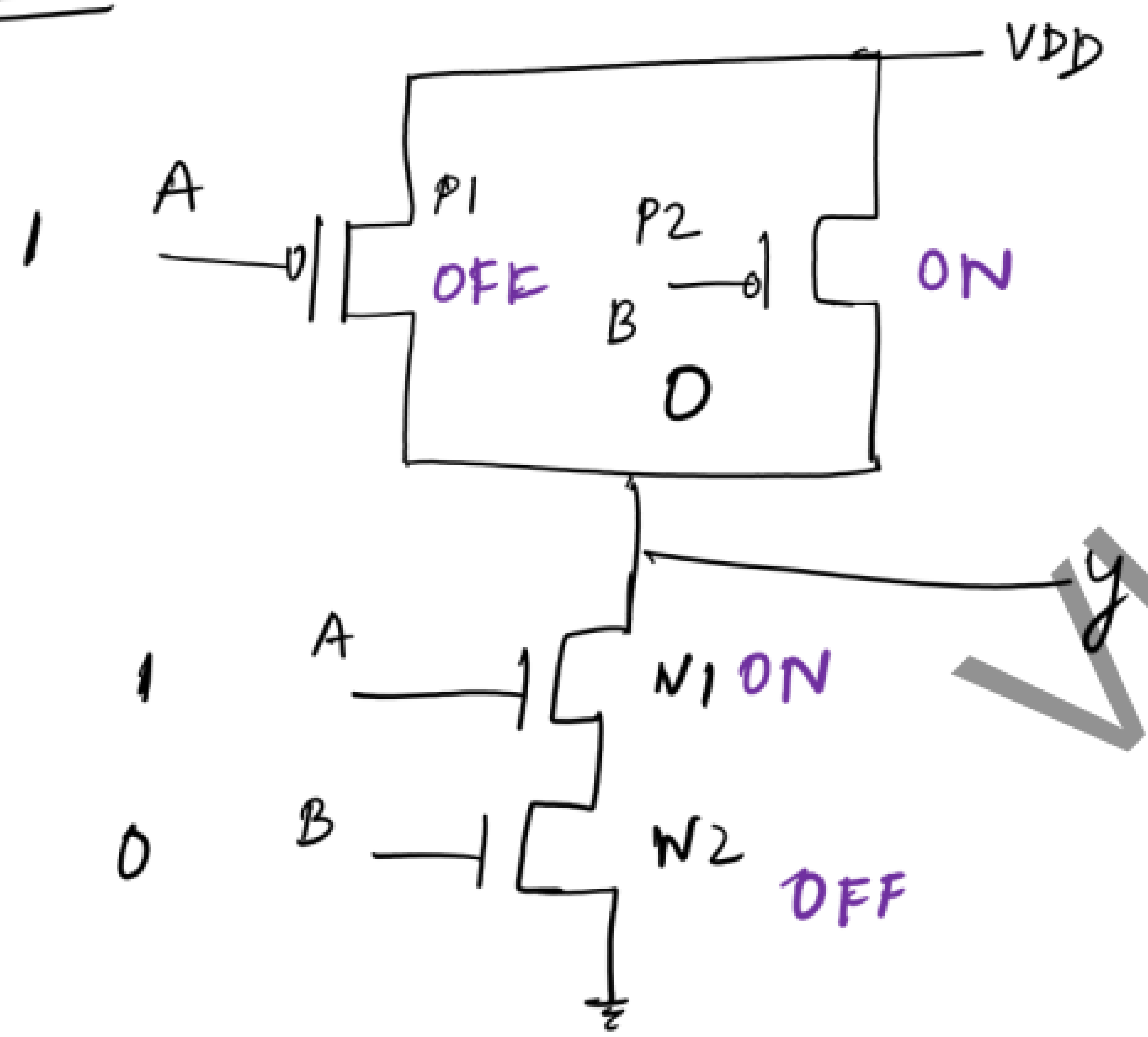
This pattern will detect N1 stuck on fault.



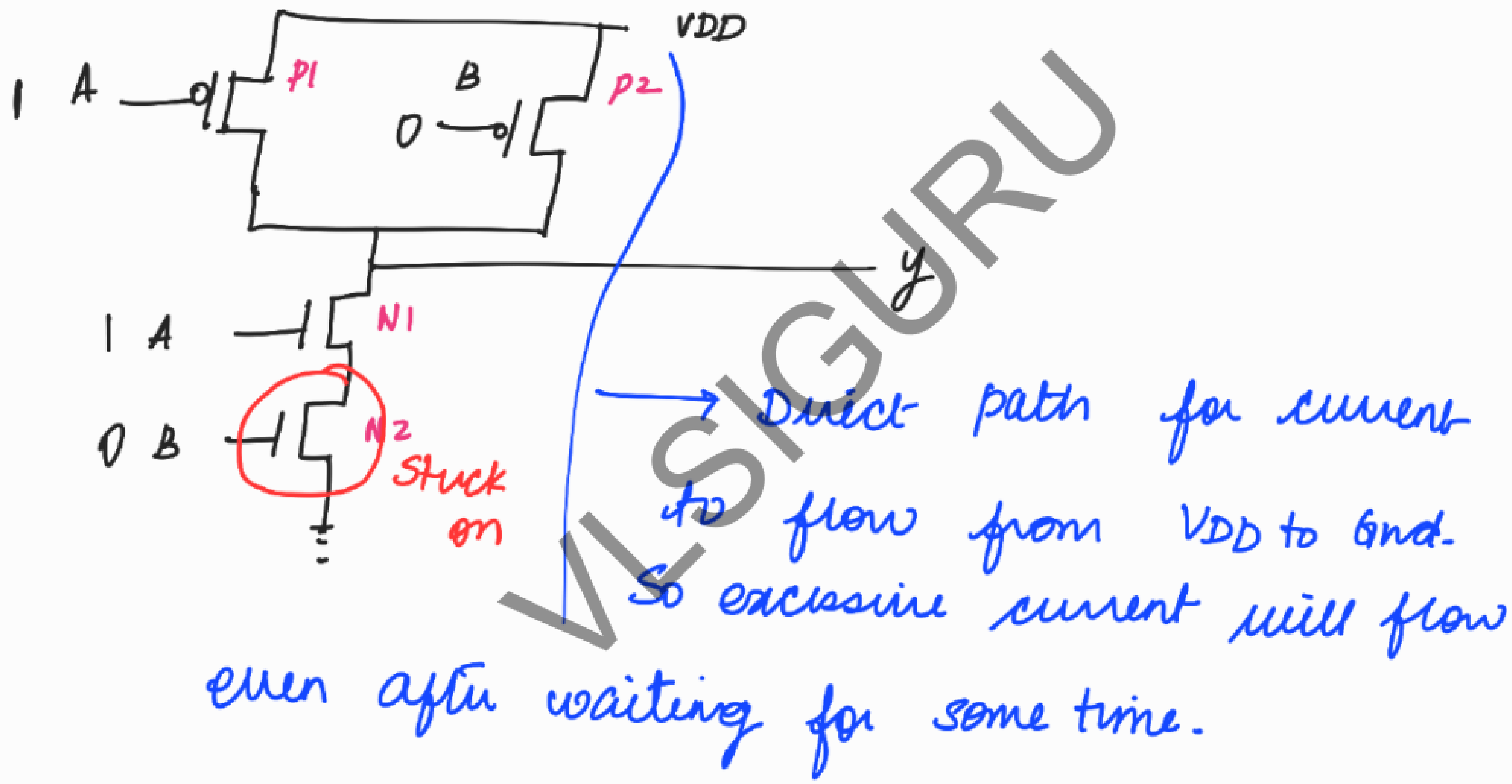
Pattern :- A = 1
B = 0

No direct path for the current to flow from VDD to Gnd.

Good :

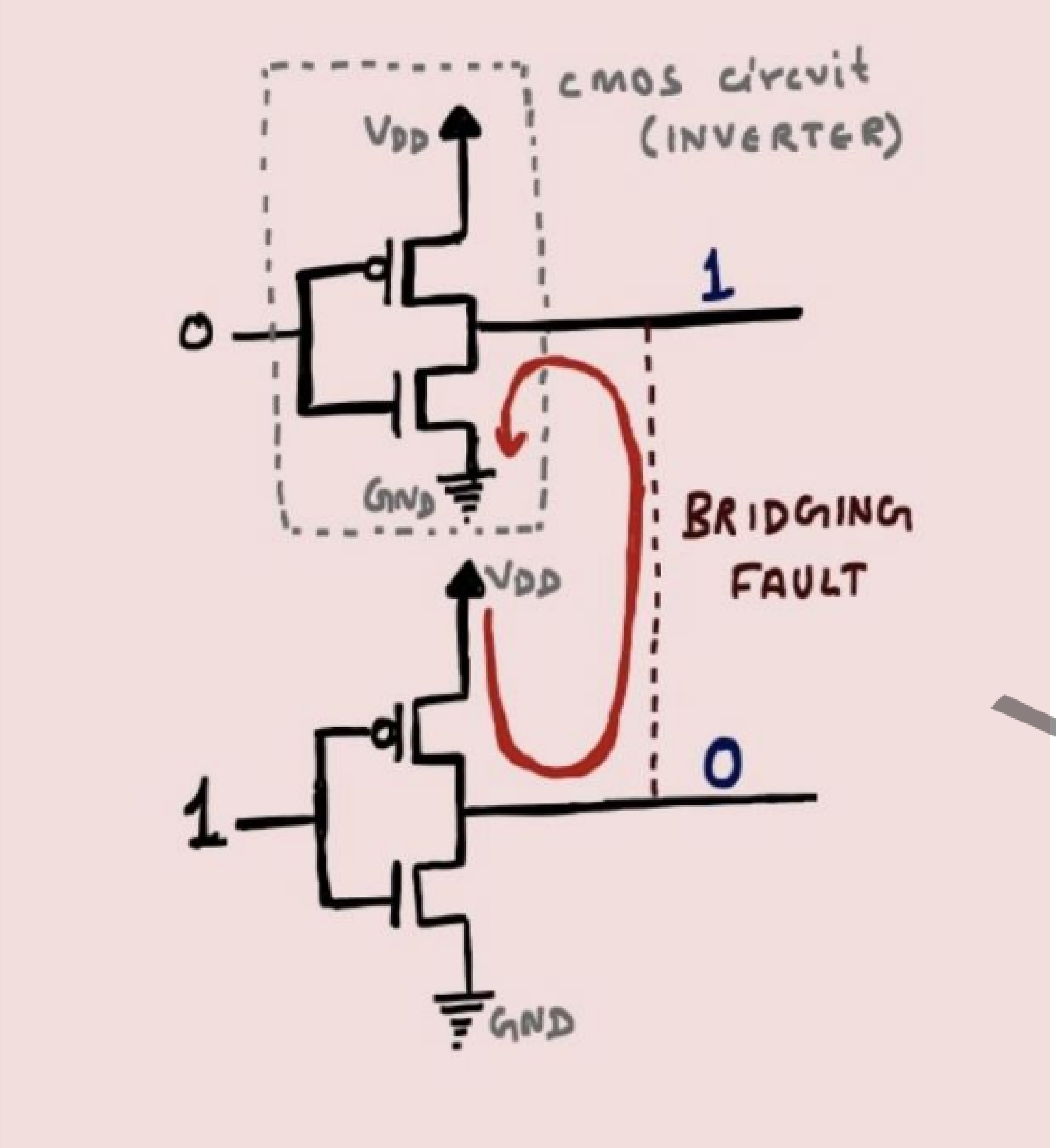


This pattern will detect N2 stuck on fault.



BRIDGING FAULTS

IDDQ tests can also detect bridging faults.



There are two CMOS circuits, where the two output wires (having opposite values) are close to each other. The bridging defect can cause a short between them, which creates a current path between VDD and GND.

NOTE

After manufacturing IDDQ current in the silicon will be compared with the value given by power analysis team (expected value).

The leakage current of each standard cell will be available in the .lib file. Power analysis team will collect all these information and give the expected

value

DC Characteristics of CMOS Inverter

https://www.youtube.com/watch?v=l2FS6lfIoW8&pp=ygUsZGMgdHJhbnNmZXIgY2hhcmFjdGVyaXN0aWNzIG9mIGNtb3MgaW52ZXJ0ZXI%3D